The Timing of Early-Life Shocks and Adolescent Mental Health

October 2026

Do the effects of an earthquake in early childhood persist?



Shock 27 February 2010 (27-F), magnitude 8.8

Exposed Children aged 6 months to 4 years

Effects of early-childhood shocks

Birth weight (Torche, 2011)

Cognitive and non-cognitive development (Gillmore, 2026)

Height and learning (Andrabi, Daniels, and Das, 2023)

Adult mental health (Persson and Rossin-Slater, 2018)

This paper

Mental health in adolescence, ages 14–18

▸ PGA map

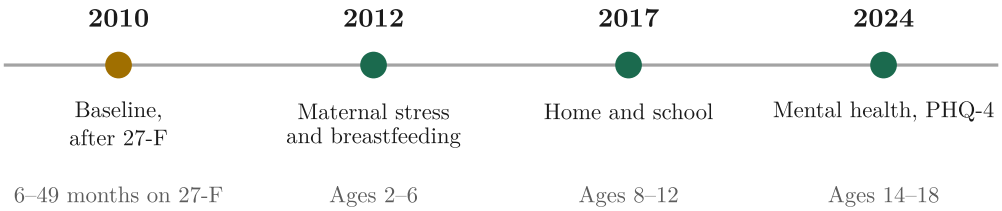
Does it matter at what age children live through the shock?

"The effects of stress on the brain depend critically on the timing (age of onset and duration). When stress occurs early in life it can have profound and lasting effects on brain organization and function." (Gee and Casey, 2015, p. 184)

Potential mechanisms

	Channels
Socioeconomic and family environment	The earthquake reduces income, damages housing, and increases conflict at home (Conger and Donnellan, 2007; Almond, Currie, and Duque, 2018)
Transmission of maternal stress	Infants regulate their stress through the mother. If she is stressed, she passes it on through daily contact (Persson and Rossin-Slater, 2018)

Data



Survey	ELPI, Early Childhood Longitudinal Survey, Chile
Sample	9,996 adolescents born between January 2006 and August 2009, aged 6 to 49 months on 27-F; 4,961 girls
Respondents	The caregiver in every round; the child in 2017 and 2024

▸ Descriptives

Econometric specification

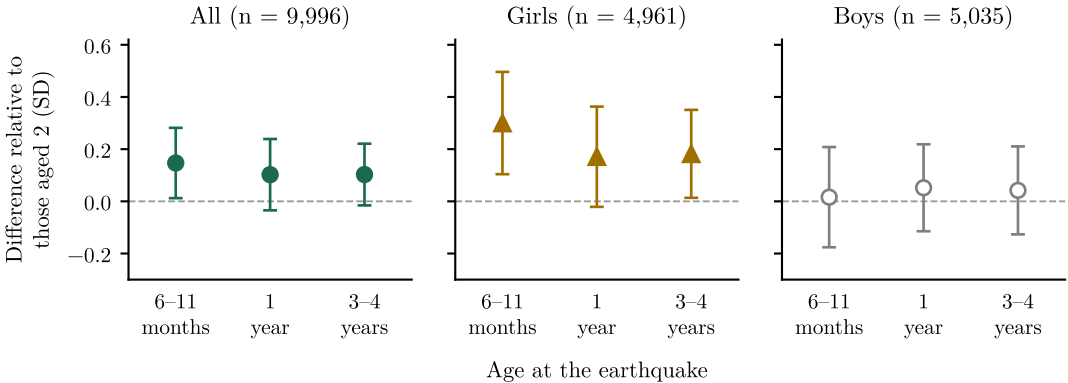
$$Y_{imc} = \sum_{a \neq 2} \beta_a \left(\text{Age}_{ai} \times \text{Affected}_m \right) + \sum_{a \neq 2} \gamma_a \, \text{Age}_{ai} + \psi_m + \theta_c + X_i' \delta + \varepsilon_{imc}$$

Y_{imc}	Mental health at ages 14–18, PHQ-4
Age_{ai}	Age at the earthquake: 6–11 months, 1 year, or 3–4 years; reference, 2 years
Affected_m	Municipality in the zone of destructive damage
ψ_m, θ_c	Municipality and birth-year fixed effects
X_i	Linear trend in month of birth; child and household controls

The earthquake date is the same for everyone, so age at the earthquake is a function of the month of birth. Replacing γ_a, θ_c, and the trend with month-of-birth fixed effects leaves the estimates unchanged (Table A4).

▸ Balance ▸ Attrition

Effect on mental health

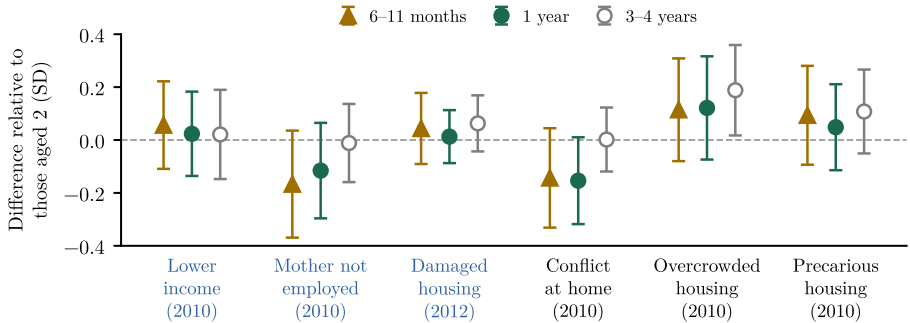


Mental health is measured with the PHQ-4 (Kroenke et al., 2009), four questions answered by the adolescent on symptoms of anxiety and depression (nervousness, worry, loss of interest, and feeling down). Each point is the difference between affected and unaffected municipalities, with 95% CI.

▸ PHQ-4 distribution ▸ Table

Socioeconomic and family environment does not explain the gap

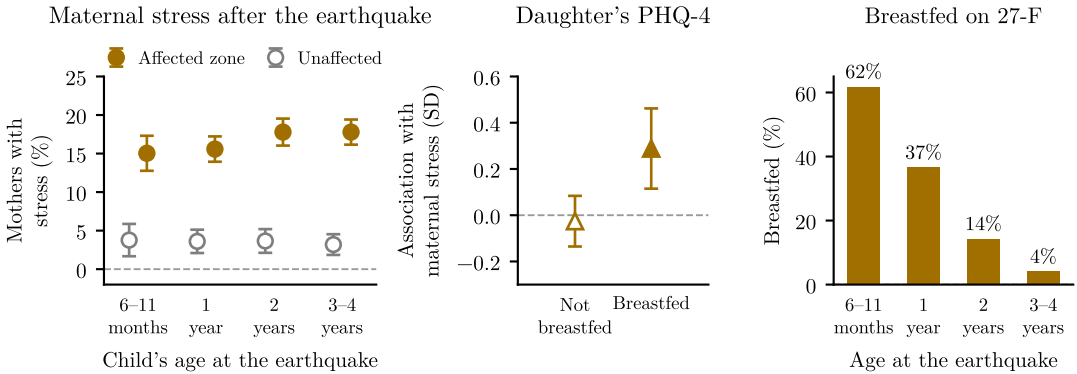
The earthquake worsened the environment (Gillmore, 2026), but with no differences by age



Measured in 2010, one to nine months after the earthquake, except housing damage, reported in 2012. In blue, the mechanisms in Gillmore (2026). Lower income, per capita; damaged housing, destroyed or with major damage; conflict at home, the mother shouting, being hostile, or hitting during the visit; overcrowded housing, persons per bedroom; precarious housing, an emergency shack (*mediagua*) or a lent or shared dwelling. Girls, relative to those aged 2; 95% CI.

▸ Table

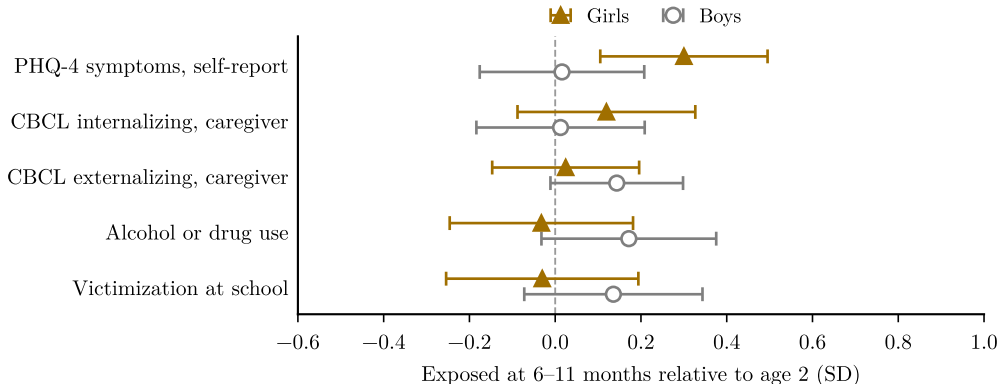
Maternal stress and daughters' mental health by breastfeeding



Active breastfeeding on 27-F serves as a proxy for regulatory dependence and biochemical and behavioral proximity to the mother. Girls, with fixed effects and controls, also for the mother's education and for age; 95% CI.

▸ Table ▸ Means

Among boys, only weak signs in behavior and substance use



▸ Table, girls ▸ Table, boys

Findings

- Girls exposed at 6–11 months have worse mental health than those exposed at age 2, by 0.30 SD on the PHQ-4.
- The socioeconomic and family environment changed similarly at all ages and does not explain this difference.
- Maternal stress is associated with worse mental health only among daughters who were breastfed, 62% of those exposed at 6–11 months.

Open questions

- What explains the effect at ages 3–4, if it operates neither through the socioeconomic and family environment nor through maternal stress?

  Hypothesis: at that age children can already remember consciously what they experience, and the memory of the earthquake could affect their mental health.

References

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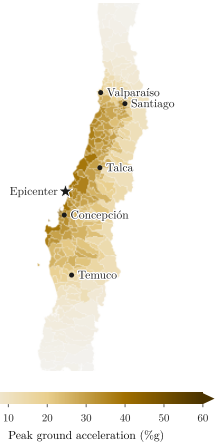
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Figure A1—Peak Ground Acceleration by Municipality



Notes: Peak ground acceleration of the 27-F earthquake, in percent of gravity, at the centroid of each municipality. Affected municipalities in the main specification are those of the six regions with destructive shaking; the map shows how shaking varied within and across them. *Source:* USGS ShakeMap of the event.

Back

Table A1—Descriptive Statistics

	All		Mean by sex	
	(1) Mean	(2) SD	(3) Girls	(4) Boys
Panel A. Adolescent, 2024				
PHQ-4 score (0–12)	4.32	2.68	4.89	3.78
PHQ-4 of 6 or more (share)	0.27	0.44	0.34	0.19
Age in August 2024 (years)	16.76	1.05	16.76	16.76
Age at the earthquake (months)	27.06	12.63	27.07	27.06
Affected municipality (share)	0.78	0.42	0.78	0.78
Panel B. Birth and household, 2010				
Birth weight (kg)	3.39	0.50	3.34	3.44
Gestation (weeks)	38.64	2.01	38.69	38.59
Mother's age (years)	29.52	7.11	29.60	29.44
Mother's education (years)	10.31	3.18	10.28	10.34
Household size	4.93	1.68	4.92	4.93
Rural (share)	0.09	0.29	0.09	0.09
Panel C. After the earthquake, 2012				
Breastfed on 27-F (share)	0.24	0.43	0.24	0.24
Mother reported stress (share)	0.14	0.34	0.14	0.13
Adolescents	9,996		4,961	5,035

Notes: Adolescents with a PHQ-4 score in 2024 (9,996 of 10,003), weighted with ELPI sampling weights. Panel C was reported in 2012; breastfed means still breastfed on 27-F.

◂ Back

Table A2—Balance on Characteristics Fixed Before the Earthquake

	At birth (SD)			Mother, 2010 (years)	
	(1) Weight	(2) Length	(3) Gestation	(4) Education	(5) Age
6–11 months × Affected	0.193	0.036	0.110	−0.560	−0.129
	(0.118)	(0.096)	(0.111)	(0.333)	(0.775)
1 year × Affected	0.105	0.031	0.096	−0.481	−0.139
	(0.103)	(0.100)	(0.086)	(0.264)	(0.695)
3–4 years × Affected	0.245	0.172	0.182	−0.350	0.457
	(0.083)	(0.099)	(0.075)	(0.300)	(0.603)
Observations	4,530	4,525	4,818	4,882	4,920
Municipality FE	Yes	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	Yes	Yes
Age-group FE	Yes	Yes	Yes	Yes	Yes
Controls	No	No	No	No	No

Notes: Girls. Each column regresses a characteristic fixed before the earthquake on the age group × affected interactions, with municipality, birth-year, and age-group fixed effects and no controls. Birth weight, length, and gestation are standardized; the mother's education and age were reported in 2010. Adding the three birth outcomes as controls, the PHQ-4 coefficient for girls exposed at 6–11 months is 0.268 (0.100), against 0.300 (0.100) in Table A4 (4,190 girls with birth records). Standard errors clustered by municipality of selection (116 clusters) in parentheses.

Back

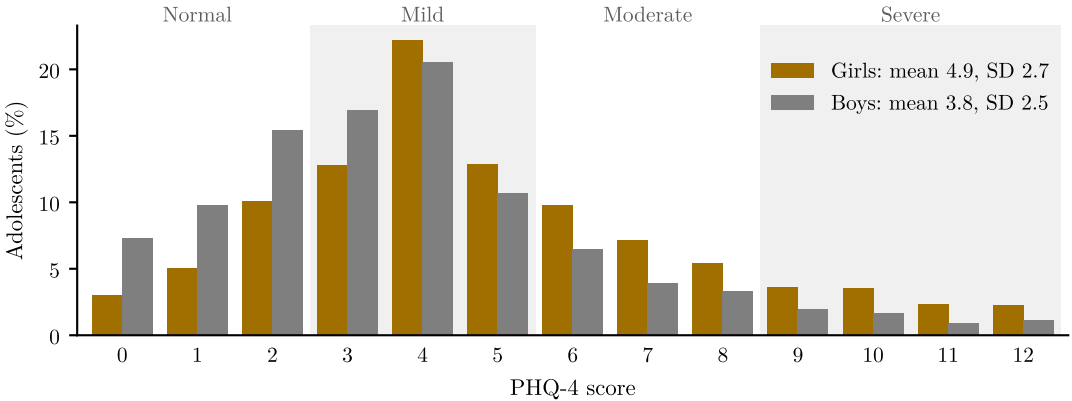
Table A3—Attrition between 2010 and 2024

	Interviewed in 2024			Birth weight, girls (SD)	
	(1)	(2)	(3)	(4)	(5)
	Girls	Boys	All	2010 baseline	Interviewed in 2024
6–11 months × Affected	−0.047	0.019	−0.019	0.064	0.120
	(0.044)	(0.043)	(0.030)	(0.098)	(0.115)
1 year × Affected	0.009	0.049	0.026	0.087	0.073
	(0.035)	(0.035)	(0.025)	(0.075)	(0.092)
3–4 years × Affected	0.019	0.028	0.019	0.145	0.189
	(0.035)	(0.034)	(0.024)	(0.076)	(0.093)
Joint test, p-value	0.47	0.57	0.43	0.30	0.24
Mean of dependent variable	0.67	0.67	0.67		
Observations	6,990	7,171	14,161	6,409	4,300
Region FE	Yes	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	Yes	Yes
Age-group FE	Yes	Yes	Yes	Yes	Yes

Notes: Children assessed in the 2010 baseline (14,161 of 15,175). Columns 1–3: equals one if interviewed in 2024. Columns 4–5: standardized birth weight of all girls assessed in 2010 and of those interviewed in 2024. Age at the earthquake is the age at the 2010 assessment minus the months from 27-F to the region's median assessment month, for stayers and leavers alike. The 2010 files do not identify the municipality: region fixed effects, no weights, and heteroskedasticity-robust standard errors in parentheses.

◂ Back

Figure A2—Distribution of the PHQ-4 Score



Notes: PHQ-4 score in 2024, weighted with ELPI sampling weights. The score adds four items scored 0 to 3; Kroenke et al. (2009) classify 0–2 as normal, 3–5 as mild, 6–8 as moderate, and 9–12 as severe.

Back

Table A4—Effect on Mental Health

	PHQ-4 (SD)			
	(1) All	(2) Girls	(3) Boys	(4) Girls
6–11 months × Affected	0.147	0.300	0.016	0.310
	(0.069)	(0.100)	(0.098)	(0.101)
1 year × Affected	0.102	0.171	0.052	0.178
	(0.070)	(0.098)	(0.085)	(0.099)
3–4 years × Affected	0.103	0.182	0.042	0.177
	(0.060)	(0.086)	(0.086)	(0.089)
Observations	9,996	4,961	5,035	4,961
Municipality FE	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	No
Age-group FE	Yes	Yes	Yes	No
Month-of-birth FE	No	No	No	Yes
Controls	Yes	Yes	Yes	Yes

Notes: Difference between affected and unaffected municipalities in the standardized PHQ-4 for each age at the earthquake, relative to those aged 2. Controls: a linear trend in month of birth, sex, the mother's age and education, household size, and rural residence (2010), the mother's number of children, and family history of mental illness (2012). Column 4 replaces the birth-year and age-group fixed effects and the trend with month-of-birth fixed effects. ELPI sampling weights. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

◂ Back

Table A5—The Socioeconomic and Family Environment

	Mechanisms in Gillmore (2026), SD			Other household conditions, SD		
	(1) Lower income	(2) Mother not employed	(3) Damaged housing	(4) Conflict at home	(5) Overcrowded housing	(6) Precarious housing
6–11 months × Affected	0.056	−0.167	0.044	−0.143	0.114	0.094
	(0.084)	(0.103)	(0.069)	(0.096)	(0.099)	(0.095)
1 year × Affected	0.024	−0.116	0.013	−0.154	0.121	0.048
	(0.081)	(0.092)	(0.051)	(0.084)	(0.100)	(0.083)
3–4 years × Affected	0.021	−0.011	0.063	0.002	0.188	0.108
	(0.086)	(0.075)	(0.054)	(0.062)	(0.087)	(0.081)
Observations	4,478	4,925	4,423	4,701	4,961	4,960
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Age-group FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Girls. Each column is a standardized index of the household environment; higher is worse. Measured in 2010, one to nine months after the earthquake, except housing damage, reported in 2012. Lower income, per capita; damaged housing, destroyed or with major damage; conflict at home, the mother shouting, being hostile, or hitting during the visit; overcrowded housing, persons per bedroom; precarious housing, an emergency shack or a lent or shared dwelling. Specification as in Table A4. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

◂ Back

Table A6—Maternal Stress, Breastfeeding, and Daughters' Mental Health

	Daughters' PHQ-4 (SD)		
	(1) Baseline	(2) Mother's SES	(3) Mother's SES and infancy
Maternal stress × not breastfed	−0.026	−0.005	−0.004
	(0.056)	(0.063)	(0.064)
Maternal stress × breastfed	0.288	0.280	0.286
	(0.089)	(0.096)	(0.112)
Difference, breastfed − not breastfed	0.314	0.285	0.290
	(0.108)	(0.102)	(0.110)
Observations	4,409	4,377	4,377
Municipality FE	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes
Age-group FE	Yes	Yes	Yes
Age group × breastfed FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Stress and breastfed × mother's education and employment	No	Yes	Yes
Stress × 6–11 months	No	No	Yes

Notes: Girls. Maternal stress equals one if the mother reported stress, distress, or anxiety after the earthquake (2012); breastfed equals one if the child was still breastfed on 27-F (2012 report). Column 1 is the estimate in the figure; column 2 lets stress and breastfeeding vary with the mother's education and employment (2010); column 3 also lets stress vary at 6–11 months. Controls and weights as in Table A4. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

◂ Back

Table A7—Maternal Stress and Breastfeeding by Age at the Earthquake

	Mothers with stress (%)		Breastfed (%)
	(1) Affected	(2) Unaffected	(3) On 27-F
6–11 months	15.0	3.8	61.8
	(1.2)	(1.1)	
1 year	15.6	3.6	36.6
	(0.8)	(0.8)	
2 years	17.8	3.7	14.3
	(0.9)	(0.8)	
3–4 years	17.8	3.2	4.1
	(0.8)	(0.7)	
Observations	6,772	2,132	8,874

Notes: Unweighted means, with standard errors in parentheses, by the child's age at the earthquake, among the adolescents of Table A1 whose mother answered each question in 2012. Columns 1 and 2 report the share of mothers who reported stress, distress, or anxiety after the earthquake, in affected and unaffected municipalities; column 3, the share of children still breastfed on 27-F.

◂ Back

Table A8—Other Outcomes, Girls

	Adolescent report (SD)			Caregiver report, CBCL (SD)	
	(1) PHQ-4	(2) Substance use	(3) Victimization	(4) Internalizing	(5) Externalizing
6–11 months × Affected	0.300	−0.032	−0.030	0.119	0.024
	(0.100)	(0.109)	(0.114)	(0.106)	(0.087)
1 year × Affected	0.171	0.026	0.044	0.154	0.100
	(0.098)	(0.096)	(0.081)	(0.120)	(0.107)
3–4 years × Affected	0.182	−0.094	0.081	0.042	−0.082
	(0.086)	(0.083)	(0.092)	(0.095)	(0.083)
Observations	4,961	4,961	4,539	4,748	4,748
Municipality FE	Yes	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	Yes	Yes
Age-group FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: Girls. Each column is a separate regression with the specification of Table A4. The adolescent reports the PHQ-4, substance use (any alcohol, tobacco, cannabis, or other drug in the last 12 months), and peer victimization and class climate; the caregiver reports internalizing and externalizing problems (CBCL). All measured in 2024 and standardized; higher is worse. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

◂ Back

Table A9—Other Outcomes, Boys

	Adolescent report (SD)			Caregiver report, CBCL (SD)	
	(1) PHQ-4	(2) Substance use	(3) Victimization	(4) Internalizing	(5) Externalizing
6–11 months × Affected	0.016	0.172	0.136	0.012	0.144
	(0.098)	(0.104)	(0.106)	(0.100)	(0.079)
1 year × Affected	0.052	−0.064	0.112	−0.063	0.053
	(0.085)	(0.089)	(0.086)	(0.099)	(0.092)
3–4 years × Affected	0.042	−0.031	0.098	0.010	0.126
	(0.086)	(0.079)	(0.094)	(0.099)	(0.090)
Observations	5,035	5,034	4,637	4,804	4,804
Municipality FE	Yes	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	Yes	Yes
Age-group FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: Boys. Outcomes and specification as in Table A8. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

◂ Back